

M.Tech (AIML/DSE) – Machine Learning

Assignment – 2: Classification Models and Streamlit Deployment

Dataset: Breast Cancer Wisconsin (Diagnostic) – UCI ML Repository

569 instances, 30 predictive continuous features; binary target: malignant/benign.

Submission links – replace these placeholders before final submission:

GitHub Repository: PASTE YOUR GITHUB REPOSITORY LINK HERE

Live Streamlit App: PASTE YOUR STREAMLIT APP LINK HERE

BITS Virtual Lab Screenshot: INSERT THE REQUIRED SCREENSHOT HERE

Problem Statement

Implement multiple classification models on one public classification dataset, evaluate them using Accuracy, AUC, Precision, Recall, F1 Score and MCC, and demonstrate the models using an interactive Streamlit application.

Dataset Description

The Breast Cancer Wisconsin (Diagnostic) dataset from the UCI Machine Learning Repository contains 569 instances and 30 continuous predictive features. The features describe cell nuclei characteristics computed from digitized fine needle aspirate images. The binary target is malignant (M) or benign (B). The ID field is excluded from modeling.

Model Comparison

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9649	0.9960	0.9750	0.9286	0.9512	0.9245
Decision Tree	0.9298	0.9246	0.9048	0.9048	0.9048	0.8492
kNN	0.9561	0.9823	0.9744	0.9048	0.9383	0.9058
Naive Bayes	0.9386	0.9934	1.0000	0.8333	0.9091	0.8715
Random Forest	0.9649	0.9942	1.0000	0.9048	0.9500	0.9258

Model Observations

Logistic Regression: Excellent overall performance with 96.49% accuracy and 0.9960 AUC. It gives a strong balance between precision and recall and is highly suitable for this dataset.

Decision Tree: The lowest overall performance among the five models in this experiment. Its 92.98% accuracy and 0.9246 AUC indicate that a single tree has higher variance and is less stable than the ensemble approach.

kNN: Strong performance with 95.61% accuracy and 0.9823 AUC. Standardization is important because kNN is distance-based. It achieves good recall while maintaining high precision.

Naive Bayes: Good AUC of 0.9934 and 93.86% accuracy. Its precision for the malignant class is 1.0000 in this test split, but recall is lower at 0.8333, so it misses some malignant cases.

Random Forest: Excellent performance with 96.49% accuracy, 0.9942 AUC and 1.0000 precision. It has the highest MCC (0.9258) in this split and is competitive with Logistic Regression.

Streamlit Features

The application supports CSV test-data upload, model selection, display of the six required metrics, confusion matrix, classification report, prediction results, and CSV download.

Deployment Checklist

- Upload app.py, requirements.txt, README.md, test_data.csv and model files to GitHub.
- Deploy app.py using Streamlit Community Cloud.
- Test the live app using test_data.csv.
- Replace both link placeholders with actual clickable links.
- Insert the BITS Virtual Lab screenshot.
- Submit the final single PDF before the stated deadline.